

Medical Software Development

Master Medical Informatics

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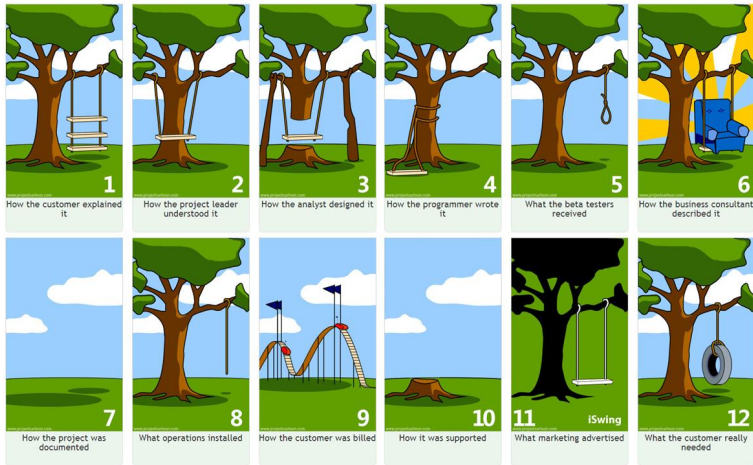


Figure: A tree swing story

ISO/IEC/IEEE 29148-2011 *Systems and Software Engineering - Life Cycle Processes - Requirements Engineering*

*...is an **interdisciplinary** process concerned with discovering, eliciting, developing, analyzing, determining verification methods, validating, communicating, documenting, and managing requirements.*

FDA - General Principles of Software Validation

A documented software requirements specification provides a baseline for both validation and verification. The software validation process cannot be completed without an established software requirements specification.

Requirement: statement which translates or expresses a need and its associated constraints and conditions to solve a problem or to achieve an objective.

Stakeholder: individual or organization having a right, share, claim, or interest in a system or in its possession of characteristics that meet their needs and expectations. Stakeholders include, but are not limited to, end users, end user organizations, supporters, developers, producers, trainers, maintainers, disposers, acquirers, customers, operators, supplier organizations, accreditors, and regulatory bodies.

- Elicitation:** build an understanding of the problem that the software is required to solve. It is fundamentally a human activity and is where the stakeholders are identified and relationships established between the development team and the customer.
- Analysis:** detect and resolve conflicts between requirements, discover the bounds of the software and how it must interact with its organizational and operational environment, elaborate system requirements to derive software requirements.
- Architectural Design:** the system structure with its components, their connections and interfaces.

Specification: create a document that can be systematically reviewed, evaluated, and approved.

- ▶ Stakeholder Requirements Specification
- ▶ System Requirements Specification
- ▶ Software Requirements Specification

Validation: verify that a requirements document conforms to company standards and that it is understandable, consistent, and complete

- ▶ **Who** is involved?
(User profiles, Stakeholders, Responsibilities, Organization)
- ▶ **What** is the current state?
(Business processes, problems, weaknesses, opportunities)
- ▶ **When** must the system be operational?
(Timeline, milestones)
- ▶ **Where** will the system be installed?
(Hardware and Software environment)
- ▶ **Why** is it necessary to develop a new system?
(goals, benefits, risks)
- ▶ **What** functions and performance should the system provide?
(use cases, user stories, data size, concurrent access)
- ▶ **Which** restrictions have to be considered?
(programming languages, availability, response time, storage etc.)

- ▶ Software Requirements Specification (SRS)
A SRS document is crafted at the initial phase of a project (to kick off the project). Depending on the used development model, requirements will be converted into
 - ▶ User Stories (Agile Approach)
 - ▶ Use Cases (Unified Process)
- ▶ Project plan

Sources of Information:

- ▶ Interviews, Workshops, Scenarios, Observations
- ▶ Literature: Books, Journals, Manuals
- ▶ Documents, Forms, Instructions, Regulations
- ▶ Internet

Functional requirements describe the functions that the software is to execute.

- ▶ Inputs (Events, Triggers)
- ▶ Outputs (responses, results)
- ▶ Preconditions (additional information needed)
- ▶ Actions
- ▶ Postconditions
- ▶ Side effects

Nonfunctional requirements (NFR) describe restrictions and limitations that need to be considered for the system development:

- ▶ Technical requirements
- ▶ Requirements regarding patient safety
(maybe derived from the risk management process)

Technical requirements

- ▶ usability, error handling
- ▶ documentation (installing, maintaining, using)
- ▶ portability, compatibility (hardware, operating systems, software interfaces)
- ▶ performance, scalability (response times, concurrent users, workload)
- ▶ safety, security (access permissions, protection against damage, attacks)
- ▶ reliability, availability, maintainability

Examples: en.wikipedia.org/wiki/Non-functional_requirement

- ▶ It is a bridge between
 - ▶ The informal discovery phase of a project where the users needs are identified
 - ▶ The actual design of the software itself
- ▶ It is the master document for a project
- ▶ It captures in a structured format all the necessary information from the discovery phase of a project

The Software Requirements Specification (SRS) establishes the basis for agreement between customers and contractors or suppliers (in market-driven projects, these roles may be played by the marketing and development divisions) on what the software product is to do as well as what it is not expected to do. It should also provide a realistic basis for estimating product costs, risks, and schedules.

Organizations can also use a SRS document as the basis for developing effective verification and validation plans.

The role of this document

- ▶ Input to the rest of the project's design documents
- ▶ It is addressed to the users
Users can ensure that their needs are addressed by the design
- ▶ It clarifies formally and in writing the informal understanding arrived during the informal process

IEEE Std 830, ISO/IEC/IEEE 29148

Title

What, Who, When, Where

Content

Summary

History

1. Introduction

1.1 Purpose

of this SRS and intended audience

1.2 Scope

summary of core functions, benefits, objectives

1.3 Definitions and Acronyms

1.4 References

1.5 Overview

1.6 Significance (underlying disease, how solution will help)

2. Overall Description

2.1 Product Perspective

business case and context, architecture, deployment diagram

2.2 Product Functions

intended uses, major functional capabilities, use case diagram

2.3 User Characteristics

user classes (actors, intended users), skills and knowledge

2.4 General Constraints

standards, rules, regulations, target platforms, legacy environment

2.5 Assumptions and Dependencies

factors that impact the requirements

3. Specific Requirements

Definition of functional and nonfunctional requirements using:

- *Use Cases*
- *GUI mockups*
- *UML diagrams (classes, sequences, state-transitions ...)*
- *data flow diagrams, data catalogs*
- *Entity relationship diagrams (database model)*
- *file formats, communication protocols*